R6.2 run report: pooled multi-target rho on an isogeny class, and path-finding cost

Run id `run-20261018T2219Z` (UTC start of the batch run). **Commit** base `abee07ee73` (origin/main) plus the uncommitted R6.2 changes on branch `research/r6-pooled-rho`, landed in the same PR as this report. **Host** `vm`, 4 vCPU, shared with other build jobs: **no isolation earned**; every recorded result is a count, wall times are contended notes only. Toy instances; nothing here concerns a deployed curve and no sub-rho claim is made.

Commands (from the worktree root):

- `target/release/examples/r6_2_pooled_class_rho --mode batch --out research/isogeny_fibers_20261010/round6/r6_2_pooled_rho/batch.json` (defaults: `--rbits 28 --trials 20 --seed 1 --ls 1,4,16,64,256`)
- `… --mode paths --out …/paths.json` (defaults: `--pbits 24,28,32 --classes 6 --pairs 10 --cap 3000 --seed 7`)

Raw records: `round6/r6_2_pooled_rho/batch.json` (schema `r6_2.batch/v1`), `paths.json` (schema `r6_2.paths/v1`), `batch.log`, `paths.log`. Both runs completed; no failures.

Batch mode (measured)

Base curve E_0: $p = 11018534707$, $\#E_0 = hr$ with $h = 40$, $r = 275462251$ ($2^{28.04}$). Class: 257 curves by BFS over rational kernels, 256 edges each with a certified dual ($\hat{\varphi} \circ \varphi = [\ell]$ checked on a point of order $> 4\ell^2$ and on the generator). Edges by $\ell \in \{2, 3, 5, 7\}$: 129, 127, 0, 0. Edges dropped for lacking a rational-kernel dual: 0, 0, 248, 248 ($p \equiv 2 \pmod 5$, $p \equiv 5 \pmod 7$, so every $\ell = 5, 7$ edge is dropped; see the note on duals below). Logs verified on their own curves: 13740.

Arms per trial (same algorithm seed for B and C): **S** one key on E_0 with `rho_walk_with` (ecbench `rho.negation`); **B** L keys on $E_1 \ldots E_L$, mapped to E_0 along the duals, one pooled batch (`rho_batch_with`, ecbench `rho.multitarget_pooled`), each log mapped back and verified; **C** L keys on E_0, same pooled batch. Unit S = total counted group-operation equivalents divided by $\sqrt{r}$, including setup. Transport is reported separately in field M and I per target.

L	trials	mean S_B ± SE	mean S_C ± SE	mean S_{single} ± SE	KS ref.	$L \cdot S_{\text{single}}$ (derived)	transport per target
1	20	1.084 ± 0.153	0.843 ± 0.106	1.002 ± 0.125	0.886	1	9 M + 1 I
4	20	2.089 ± 0.107	2.285 ± 0.092	0.98 ± 0.083	1.939	3.92	13.5 M + 1.5 I
16	20	4.272 ± 0.097	4.424 ± 0.13	0.861 ± 0.1	3.969	13.78	25.9 M + 2.9 I
64	20	9.395 ± 0.093	9.266 ± 0.112	1.114 ± 0.137	7.984	71.29	54.6 M + 6.1 I
256	20	20.165 ± 0.138	20.17 ± 0.086	1.042 ± 0.12	15.992	266.69	163 M + 18.1 I

KS ref. is the Kuhn–Struik expected pooled cost on $\{P, -P\}$, $\sum_{i<L} \binom{2i}{i} 4^{-i} \sqrt{\frac{\pi}{4}}$. $L \cdot S_{\text{single}}$ is derived as $L \times$ the recorded mean of arm S (the cost of L independent solves).

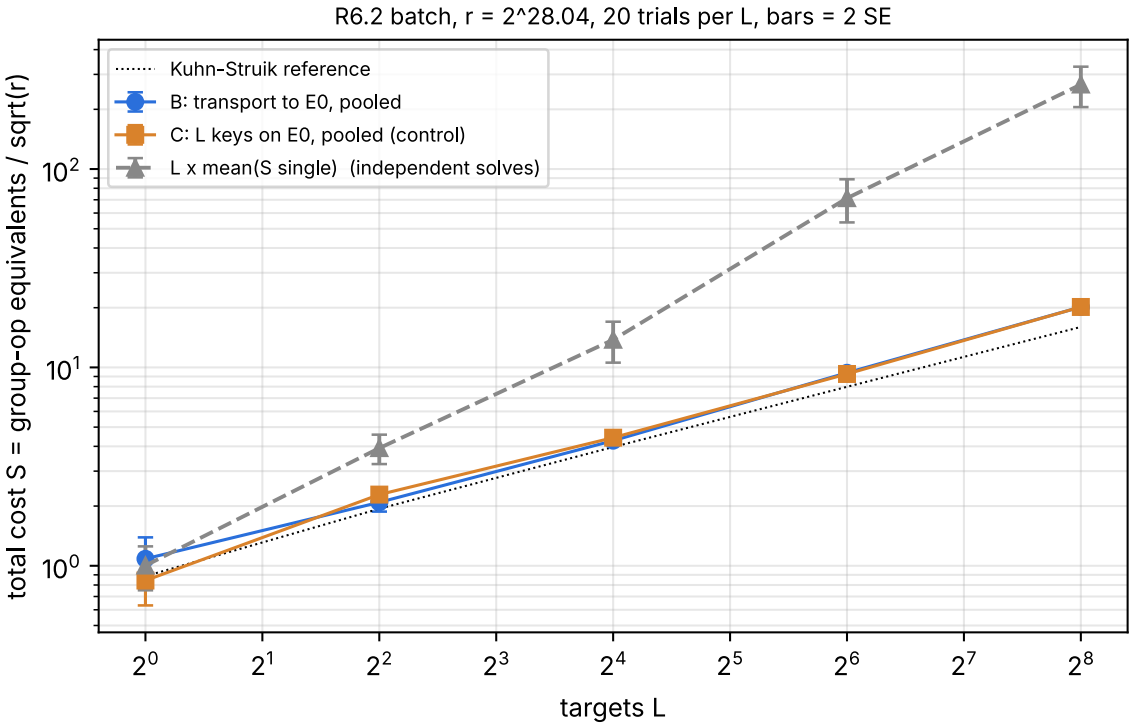


Figure 1: Batch run `run-20261018T2219Z`: recorded mean total cost per arm against L, $r = 2^{28.04}$, 20 trials per L, bars 2 SE. Source `batch.json`; plot script `plot.py`.

Observations. At every L the means of B and C lie within about 2 SE of each other. Transport costs at most 163 M + 18.1 I per target, at path depth up to 34. Both pooled arms stay above the KS reference (at $L = 256$: 20.165 and 20.17 against 15.992) and far below $L \cdot S_{\text{single}}$. No paired interval between B and C was computed by the harness, so no difference or equivalence claim is made beyond these recorded SEs.

Paths mode (measured)

For each p size, random curves until 6 components of the rational $\ell \in \{2, 3, 5, 7\}$ graph of at most 3000 curves were enumerated (components over the cap skipped and counted). For 10 random pairs per component, a bidirectional BFS found a path from scratch. Cost is counted in neighbour enumerations (one per (curve, ℓ)).

p bits	p	r	h	component	mean enum./pair	pairs
24	8937157	30181	296	480	1272	10
24	12472513	2079863	6	844	1483.2	10
24	12372967	3083	4012	240	308.8	10
24	9028927	9467	954	1356	3233.6	10
24	10800737	691	15630	2724	2003.2	10
24	13079063	131	99846	2348	5620.8	10
28	181295329	20146111	9	2424	5016	10
28	157142431	374123	420	1024	2045.6	10
28	159322511	7966141	20	8	7.6	10
28	166463293	1777	93687	692	1142.4	10
28	214809031	2147869	100	56	22	10
28	197516041	21945271	9	34	27.2	10
32	3443338453	7547	456248	30	26	10
32	2915016661	291507547	10	68	35.2	10
32	3844840117	13397	286994	10	7.2	10

32	3373723501	6438191	524	24	16	10
32	3350988913	6228671	538	620	1536	10
32	2486429123	3319727	749	647	890.4	10

Components skipped over the cap: 24 bits: 5; 28 bits: 9; 32 bits: 45.

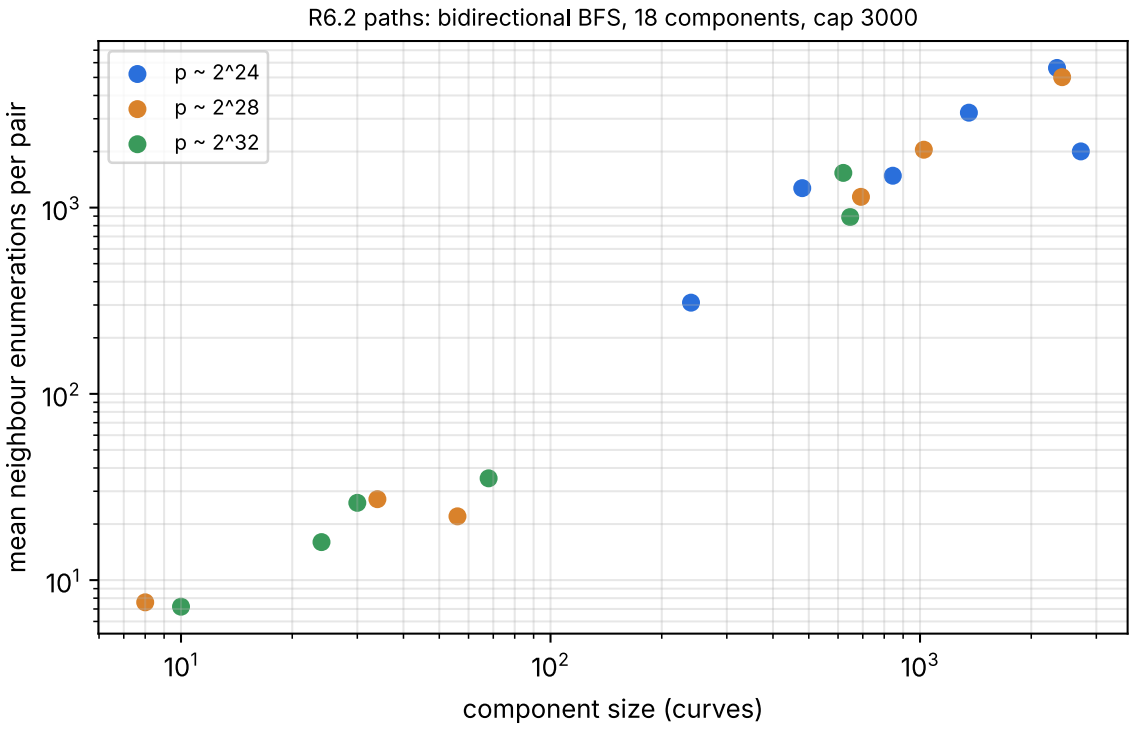


Figure 2: Paths run `run-20261018T2219Z`: recorded mean neighbour enumerations per pair against component size, 10 pairs per component. Source `paths.json`.

Observations. Mean enumerations per pair range from 7.2 to 5620.8. They grow with component size and stay on the order of the component size, here up to 2724 curves. Selection is biased toward small components by the cap; most 32-bit components exceeded it.

Note on duals for $\ell \geq 5$ (derived, test-checked)

`SmallIsogeny::dual` builds $\hat{\varphi}$ from a rational root of the ℓ-division polynomial on the codomain. If φ has rational kernel abscissas, Frobenius acts on $\ker \varphi$ by $\lambda = \pm 1$ and on $\ker \hat{\varphi}$ by $p\lambda^{-1}$, so $\hat{\varphi}$ has rational kernel abscissas iff $p \equiv \pm 1 \pmod{\ell}$. The test `degree_five_and_seven_isogenies_and_certified_duals` now asserts exactly this equivalence; it previously assumed the dual always exists and failed. The example drops such edges and records them in `edges_without_dual_by_ell`.

Missing data and limits

No isolated or independent-runner measurement; no ecbench session or paired bootstrap; one base curve for batch mode; ℓ = 5, 7 transport untested on this p.